
STUDENT & COURSE PERFORMANCE ANALYSIS

Transforming Data Into Meaningful Learning Insights

A Dual-Dashboard Insight Built from a Single Data set

NAME : P. SADHURNA

ROLE : DATA ANALYST

TOOLS USED :

- EXCEL,
- POWERBI.

DASHBOARD 1:

STUDENT PERFORMANCE OVERVIEW

DASHBOARD2 :

COURSE PERFORMANCE OVERVIEW

ABSTRACT

This project focuses on analyzing student performance, course engagement, and learning outcomes using Power BI. The dataset was first cleaned and prepared using Microsoft Excel, after which it was loaded into Power BI using the Get Data option. Two interactive dashboards were developed to identify performance gaps, weak courses, and key factors affecting student completion and satisfaction. The insights obtained from this analysis help in improving course quality and student learning experience.

OVERVIEW

With the rapid growth of online learning platforms, it is important to monitor student engagement and course effectiveness. This project provides a detailed analysis of student behavior, performance trends, and course outcomes through interactive Power BI dashboards. The analysis supports data-driven decision making for educational improvement.

PROBLEM STATEMENT

Many educational platforms face difficulty in identifying low-performing courses, poor engagement levels, and performance gaps among students. Without proper analysis, it becomes challenging to improve learning outcomes. This project aims to analyze student and course data to identify these issues and provide meaningful insights.

Data Source & Data Collection

The project uses a sample Learning Management System (LMS) data set in Excel format, covering the years 2023 to 2025. It contains student profiles, course details, assessments, and engagement-related metrics. The data set also includes feedback ratings used to analyze overall learning performance.

DATA FIELDS

- Student Name.
- Age.
- Gender.\
- Educational level
- Employment status
- Course Name.
- Course Level.
- Quiz Score.
- Project Grade.
- Progress Percentage.
- Assignments Submitted.
- Assignments Missed.
- Session Duration.
- Satisfaction Rating.
- Instructor Rating
- City.
- Device Preference.etc....

DATA PREPARATION STEPS

Step 1: Import Data into Excel

- Open Excel and go to the "Data" tab.
- Select "Get Data" and choose the appropriate import option.
- Browse to the file location and import the data.

Step 2: Clean the Data in Excel

- Check for null or blank columns/rows.
- Delete empty or unnecessary columns.
- Fix data types using Excel functions.
- Remove outliers or errors using conditional formatting.

Step 3: Analyze and Identify Problems

- Use Pivot Tables, charts, and formulas to understand the data.
- Identify trends, issues, or patterns.
- Propose solutions based on identified problems.

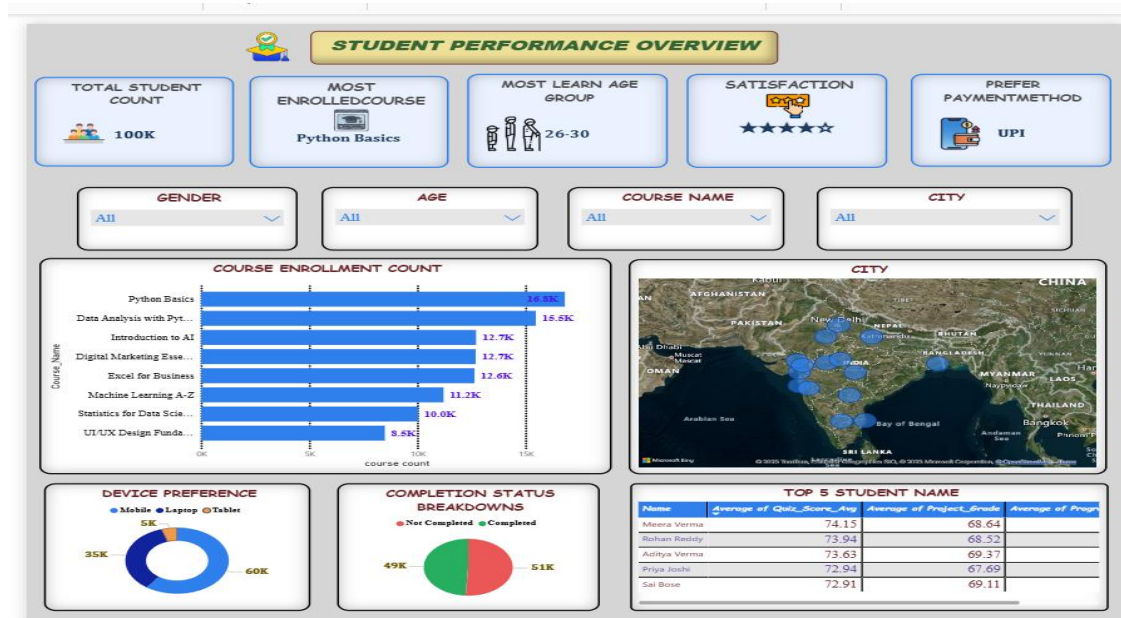
Step 4: Import Data to Power BI

- Open Power BI and go to the "Home" tab.
- Select "Get Data" and choose Excel.
- Import the cleaned Excel file.

Step 5: Create Visualizations in Power BI

- Use appropriate visuals like bar charts, pie charts, and line graphs.
- Add slicers and filters for dynamic analysis.
- Format visuals for clarity and readability.

DASHBOARD 1- STUDENT PERFORMANCE OVERVIEW



KPI's:

- ✓ Most Enrolled course - **Python basics**
- ✓ Total student count - **100K**
- ✓ Overall rating - **4.5**
- ✓ Most learning age group - **26-30**
- ✓ Most prefer payment method - **UPI**

OVERALL VISUAL ANALYZE

The visuals analyze course-wise enrollment, student engagement, completion patterns, and learning preferences. They provide insights into course popularity, geographic distribution, re-watch behavior, device usage, and top-performing students, helping to understand overall learning behavior and platform usage.

DASHBOARD 2- PERFORMANCE ANALYSIS



KPI's

- ✓ Low AssignInstructor Rating - 4
- ✓ Low Assignment Submitted Course - **UI &UX designfoundations**
- ✓ Most Assignment Submitted Course - **Python basics**

OVERALL VISUAL ANALYSIS

The visuals analyze course performance, assignment submission patterns, learning effectiveness, and performance gaps. They highlight relationships between quiz scores and progress, effort versus output, assignment submitted versus missed, and completion trends across course levels, helping identify weak areas and improvement opportunities

ANALYSIS SUMMARY & RECOMMENDATIONS

Analysis Summary

- ✓ Beginner courses show higher completion rates.
- ✓ High-enrollment courses have more missed assignments.
- ✓ Longer session duration leads to better performance.
- ✓ Instructor rating strongly impacts student satisfaction.

Recommendations

- ✓ Improve assignment engagement in popular courses.
- ✓ Provide extra support for advanced learners.
- ✓ Use feedback to enhance instructor quality.
- ✓ Refine course structure and difficulty levels.

CONCLUSION

This project demonstrates how Excel and Power BI can effectively transform raw data into meaningful insights. The dashboards offer clear visibility into student performance, engagement patterns, and course effectiveness. These insights support informed, data-driven decision-making and help enhance the overall learning experience.